

Designing Socio-Spatial Interfaces for Embodied and Situated Social Interaction

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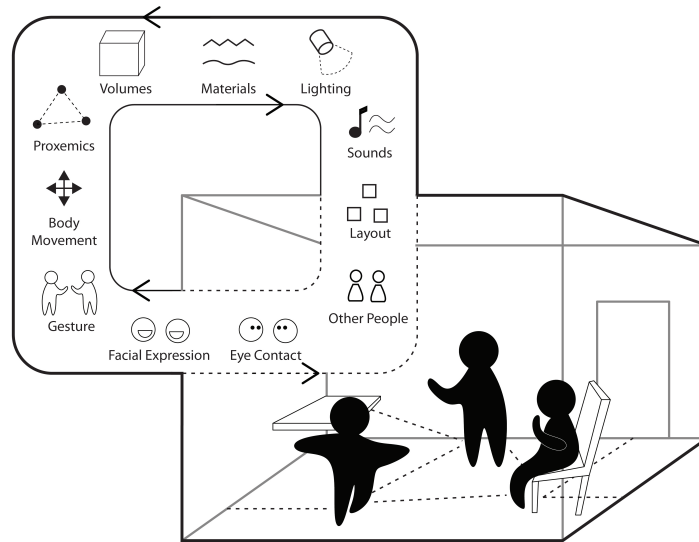


Figure 1: Socio-Spatial Interface Concept

Abstract

Physical space has a critical yet often overlooked role in fostering social interaction, particularly in HCI, where digital solutions rarely incorporate the physical environment. This dissertation aims to address this gap by exploring the design space of "socio-spatial interfaces"—interactive systems that leverage human interaction in physical environments to facilitate social interactions. Socio-spatial interfaces employ spatial mediation techniques, including projected imagery, reflections, mixed reality, and audio, to "scaffold" social interactions by integrating digital and physical environments. Through a Research through Design (RtD) approach, this dissertation investigates how these interfaces facilitate interactions among strangers through design exemplars such as *SocialStools* and *Mirror-Bot*. By drawing from architecture and social psychology, this work

aims to lay a foundation for socially adaptive environments, offering new pathways in HCI for fostering inclusivity and community through shared spaces.

CCS Concepts

• Human-centered computing → Ubiquitous and mobile computing systems and tools.

Keywords

Socio-spatial, Mixed reality, Embodied Interaction, Physical space, Environment, Social interaction

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1 Introduction

"Our body applies itself to space like a hand to an instrument, and when we wish to move about we do not move the body as we move an object. We transport it without instruments as if by magic, since it is ours and because through it we have direct access to space [26]." —

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Maurice Merleau-Ponty

The COVID-19 pandemic underscored the critical importance of in-person social interaction, a dimension of human experience that many had previously taken for granted. Research has shown that face-to-face interaction fosters empathy and deepens interpersonal connections, aspects that digital communication has struggled to fully replicate [14, 23, 39]. In response, HCI has developed a range of interventions aimed at enhancing in-person social interaction—such as conversational agents [27], wearables [17], social robots [31, 41], and public displays [15, 30, 37], etc. However, many of these solutions overlook the potential of physical space itself as a medium for facilitating social engagement.

In the field of architecture, physical space has long been recognized as a catalyst for social interaction. Designers create spaces that shape social behaviors, whether through the arrangement of furniture or the spatial layout of a room. For example, different restaurant serving hatches influence interactions between diners and cooks, ranging from brief, transactional exchanges to more engaging and participatory experiences [20]. This understanding of space as a dynamic social interface is now being integrated into HCI, where spatial environments are increasingly recognized as immersive and interactive systems. Unlike traditional HCI interfaces that follow predefined input-output structures, architectural spaces offer unmediated interaction, where elements such as furniture act as props that scaffold social engagement [20].

The integration of digital technologies into physical spaces opens up new possibilities for transforming these environments to foster interpersonal connections. Drawing from my background in architecture, I am particularly motivated by the concept that "people are inseparable from the interface itself [20]," where human behavior and spatial interaction co-create the interface itself. I explore the concept of the "socio-spatial interface," where human interaction and spatial elements co-create a seamless system. This approach blurs the distinction between human and interface, embedding embodied, contextually situated interactions within everyday environments.

This dissertation explores the design space of "socio-spatial interfaces" (Figure 1)—interactive systems that leverage human interaction within physical environments to facilitate social interactions. Recognizing that social interaction in physical space is inherently relational, shaped by spatial dynamics such as the arrangement of people and objects. This continuous loop of spatial and social dynamics constitutes the foundation of socio-spatial interfaces. Take strangers' interaction as an example, the research draws from social psychology to examine how technology-embedded space could foster togetherness between people. The socio-spatial interfaces employ spatial mediation—such as projected imagery, reflections, and audio—to scaffold social interactions by integrating digital and physical environments. With advancements in mixed reality and machine learning, the research aims to examine the embodied and situated interaction among strangers and how such interfaces could foster togetherness and community building.

2 Theoretical Foundations and Related Work**2.1 Insights from Architecture**

Socio-spatial interface design draws on architectural theories that emphasize the relationship between body, space, and movement. Henri Lefebvre's work emphasizes that the human body plays a fundamental role in shaping and defining space, arguing that space cannot be fully experienced without bodily presence [24]. This perspective aligns with the views of Yi-Fu Tuan [38], Doreen Massey [1], and Tim Ingold [16], who all emphasize that spaces are not static backdrops but dynamic environments that evolve with the activities and relationships of their occupants. Ingold, for example, contrasts wayfaring (an immersive, lived experience) with transport (goal-directed movement), suggesting that interacting with the environment allows people to build knowledge through embodied movement [16]. Together, these perspectives highlight the importance of designing interfaces that engage users' full bodies and invite meaningful, holistic interaction.

Further, Gaston Bachelard and Edward Soja add an emotional and poetic dimension to spatial design. Bachelard views spaces as repositories of memories and emotions that connect deeply to identity [2], while Soja's concept of Thirdspace blends physical and conceptual realms to create an immersive, lived experience [34]. This perspective encourages the design of socio-spatial interfaces that merge physical, digital, and emotional elements to foster reflection and creativity, ultimately enabling more profound and playful interactions within shared spaces. Additionally, the work of Deasy and Lasswell [5], while practitioner-focused, highlights connections between social theory concepts and architectural design, providing valuable insights into human behavior in designed environments. Buxton's exploration of the interplay between architectural design and interaction design [4] also underscores the potential for socio-spatial interfaces to create meaningful and contextually rich interactions.

2.2 Proxemics and F-formation

The concept of proxemics [13], introduced by Edward T. Hall, and F-formation patterns [18], described by Adam Kendon, provide insights into how physical distances and spatial arrangements affect social interactions. Proxemics explores how personal space and physical proximity influence communication and relationship dynamics. Hall's work reveals that the spatial proximity between individuals can significantly impact the nature and quality of their interactions [13]. For example, closer distances may facilitate more intimate and personal conversations, while greater distances might be suitable for formal and public interactions. F-formation patterns describe the spatial configurations people adopt during face-to-face interactions, such as circular arrangements that facilitate group discussions [18]. Kendon's insights highlight the importance of spatial arrangements in promoting effective communication and collaboration.

Further, the concept of semi-fixed features, as discussed in works such as AirConstellations [25] and Grønbaek et al.'s Proxemic Transitions [7], reveals how adaptable spatial elements can dynamically influence proxemic behaviors and interaction patterns. These insights extend the potential of designing socio-spatial interfaces

that support adaptive and contextually responsive social interactions, particularly in mixed-presence settings [35, 36]. Additionally, Krogh's work on sensitizing concepts for socio-spatial literacy [22] provides an essential framework for understanding and enhancing socio-spatial interactions in HCI.

2.3 Socio-spatial Interventions in HCI

A growing focus in HCI has been on designing socio-spatial interventions that promote embodied social interactions among strangers, leveraging physical space and user engagement to enhance communal experiences. For instance, *iFloor* allows multiple users to control floor-based graphics through their body movements, fostering collaboration and communication by making physical presence a medium for interaction [6]. Another example, *21 Balancoires*, invites people to swing together to alter background music, creating a shared, playful experience that promotes collaboration and social bonding [3]. In addition, *Robotic Bar-Stools* autonomously position users towards each other, subtly encouraging eye contact and direct engagement to facilitate more immersive social interactions [32]. Similarly, the *TouchBranch* installation explores social dynamics by examining how physical touch varies between strangers and acquaintances, offering valuable insights into how proximity and touch influence interaction patterns [19]. Lastly, *Solstice LAMP* uses light projections and ambient sound on a building façade to craft an interactive environment, encouraging passersby to experience a shared moment through synchronized visual and auditory stimuli [15].

3 Research Objective and Methods

The goal of this dissertation is to explore the design space of socio-spatial interfaces that are full-body embodied, situated in the context, and embedded in everyday physical environments, to facilitate social interaction. This research adopts a Research through Design (RtD) methodology, where the process of designing and prototyping artifacts is used as a means of generating knowledge [21, 42]. RtD is well-suited for addressing complex, "wicked" problems inherent in socio-spatial interface design. It follows an iterative, exploratory approach where prototypes are rapidly created, tested, and refined.

The research also adopts the body-storming [29] and co-design method [33]. Body-storming involves reenacting user scenarios through performance and improvisation to understand how people interact with designs in specific environments [29]. This method is especially relevant for socio-spatial interfaces, where interactions are highly context-dependent. Co-design workshops involve end-users directly in the design process, ensuring that the final product reflects their needs, preferences, and the context in which they will use the design [33]. This participatory approach could be used in designing for socio-spatial interfaces because it emphasized on the "collective creativity" of the users, instead of iterating the design based on the designer's ideology.

4 Thesis Progress To Date

4.1 Study 1: A Design Exemplar of Socio-Spatial Interface

In Study 1 [9], *SocialStools* was developed as a design exemplar of a socio-spatial interface aimed at fostering playful interactions and a sense of togetherness among strangers. As shown in Figure 2, the system consists of three stools that respond to users' physical movements through visual and auditory effects, encouraging interaction by sitting, moving, and orienting the stools relative to each other. Using projection mapping and audio feedback, *SocialStools* visualizes personal space with rippling circles, adjusts sound based on proximity, and generates visual "bubbles" reflecting users' angular orientation. This study demonstrates how embodied interactions and proxemics can create environments that invite social engagement among strangers.

4.2 Study 2: Evaluating Socio-Spatial Interface for Fostering Togetherness Across Strangers

Study 2 [8] involved an exploratory lab study with 36 participants across 12 groups, evaluating *SocialStools* for its ability to foster togetherness among strangers. Using qualitative methods to analyze behaviors such as verbal communication, non-verbal cues (e.g., eye contact, laughter), and synchronized movements, we identified four group behaviors: "Game Players" who focused on interacting with the system, "Explorers" who shifted between engaging with the system and other people, "Socializers" who primarily focused on each other, and "Solitary" participants who engaged quietly with the design. These findings suggest that socio-spatial interfaces serve as flexible social mediators, either becoming the focal point of interaction, or fading into the background when participants shift their attention to each other.

4.3 Study 3: Establishing An Analytical Framework

Through the prior studies, we identified a need for a framework to understand how stranger interactions unfold in socio-spatial settings. Drawing from social psychology and communication theories, we developed a framework of human behavior with distinct, interconnected components. This framework was applied to analyze the interactions of 81 strangers using *SocialStools* in a field study. A key finding was the emergence of "dynamic proxemics," characterized by embodied, shifting proximities between strangers. Behaviors such as claiming personal space, exploring others' personal boundaries, relaxing proxemics with laughter, and synchronizing movements as a group reveal how bodily relationships act as vehicles for social mediation in physical spaces.

4.4 Other Studies on Space Transformation

In exploring space transformation, I have conducted three additional first-authored studies, all currently under review:

(1) Co-designing with neurodivergent individuals for sensory mitigation, Project *DentAR* (Figure 3 - d) [11]: Neurodivergent individuals often experience heightened sensory sensitivity and discomfort in environments like dental clinics. Through 13 co-design sessions with neurodivergent participants, I investigate how mixed reality

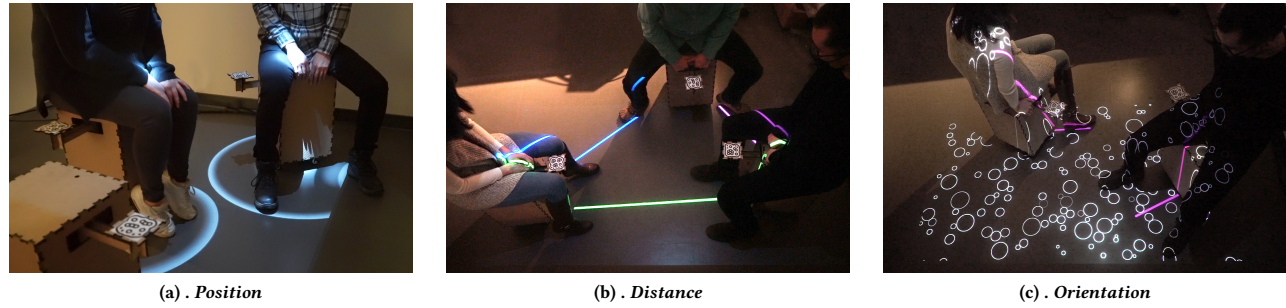


Figure 2: Three interaction setups of *SocialStools*: (a) *Position*: a visualization of personal space according to the position of the stool; (b) *Distance*: changing volume, pitch, and timbre of music output by distance between the stools; (c) *Orientation*: changing of shapes and colors of bubbles on the floor by angular orientation of the stools.

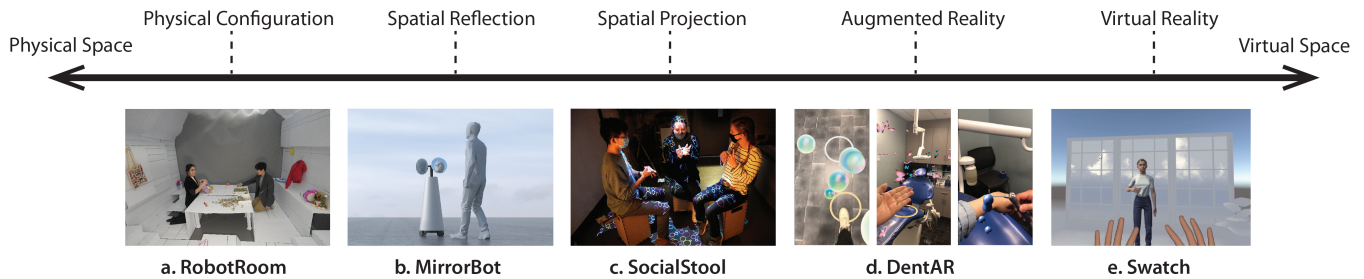


Figure 3: Studies on a spectrum from physical space to virtual space.

can be leveraged to design future interventions that mitigate sensory discomfort, transforming the clinical environment.

(2) Room-scale agile physical prototyping, Project *RobotRoom* (Figure 3 - a): While XR offers significant potential to transform the experience of physical space, they often lack the tactile and physical feedback essential for a truly embodied experience. To address this, I developed a novel approach using reconfigurable room-scale prototypes, examining how these can adapt to various daily activities such as work, socializing, entertainment, and rest. In co-design sessions with 30 participants, room-scale agile prototypes were used to envision adaptable home spaces.

(3) Spatial transformation in VR environment, Project *Swatch* (Figure 3 - e): Through three consecutive studies from an online survey to an in-person VR experiment, I examined how representations in virtual spaces can shape social dynamics.

5 Next Step 1 - Creating Adaptive Socio-Spatial Interfaces

Building on insights from my previous studies, the next step in my research is to conceptualize the design space and features of socio-spatial interfaces, exploring how this new lens can reshape our understanding of existing projects. Using the "Research through Design" approach, I plan to develop another socio-spatial interface design exemplar inspired by architectural design—specifically,

the mirror. Mirrors, often used to reflect and alter spatial perceptions, could also facilitate social interactions by creating moments of serendipitous eye contact. With this idea, I propose designing, developing, and evaluating *MirrorBot* [10] (Figure 3 - b), a moving robotic mirror system that intelligently rotates to encourage social interactions. To develop its form and movement, I will conduct body-storming workshops and then run an in-lab user study where strangers, unaware of the true study purpose, are observed to see how they naturally react to the robot and each other.

Given the complexity of human social behaviors, my second study aims to explore whether *MirrorBot* can adapt to social interactions. I propose incorporating reinforcement learning (RL) into the system to allow it to learn how to facilitate social interaction [12], as rule-based systems may be inadequate for capturing the complexity of these interactions. RL is well-suited for sequential decision-making tasks like social mediation, and it could reveal strategies beyond human heuristics. The study will involve first training the RL model in a simulated environment, deploying it in the real world, and finally conducting a user study to evaluate participants' responses to an adaptive socio-spatial interface.

6 Next Step 2 - Synthesize a Socio-Spatial Interfaces Framework

At the same time, I would like to synthesize a framework of socio-spatial interfaces. Key inquiries include: How to define socio-spatial interfaces? What are the key features of socio-spatial interfaces? For example, drawing from cognitive science, one key feature of a socio-spatial interface could be "Distributed and Expressive Representation." As the interfaces distribute cognition across people and their surroundings, they employ digital and physical representations that intuitively link actions and outcomes [28, 40]. Another possible quality is "People are Part of the Interface Itself." In socio-spatial interfaces, users are integral to the system, co-creating and reshaping it in real-time. Unlike conventional HCI, where input-output channels are distinct, socio-spatial interfaces dynamically adapt as users interact, turning people into active components of the environment [20].

However, a cohesive framework needs to be developed and iterated upon as the theoretical foundation of this dissertation. This work will involve not only identifying these potential qualities but also testing and refining them through empirical studies and iterative design. By keeping the exploration open-ended, I aim to uncover the nuances and complexities of socio-spatial interfaces, ensuring the resulting framework is both robust and adaptable.

7 Dissertation status and long-term goals

I am currently a 4th-year PhD student in Information Science at Cornell University, expecting to defend my dissertation in the summer of 2026. To date, I have published two thesis-related papers, with four additional full papers under review. I look forward to receiving feedback from the CHI '25 Doctoral Consortium to refine my thesis and am open to exploring new directions based on that input. I have not yet participated in a Doctoral Consortium, and I see this as a valuable opportunity to exchange ideas and insights, particularly in bridging spatial design knowledge with the HCI field. My long-term academic goal is to continue advancing the integration of architecture with HCI by exploring socio-spatial interfaces that foster genuine interpersonal connections, inclusivity, community building, and overall human well-being.

References

- [1] Ben Anderson. 2008. For Space (2005): Doreen Massey. *Key texts in human geography* 8 (2008), 227–235.
- [2] Gaston Bachelard. 1994. *The Poetics of Space*. Beacon Press. Original work published 1958.
- [3] Joshua Bloom. 2016. Musical Swings: An exercise in collaboration. Impacts of a temporary installation on public spaces in three US cities — A comparative analysis. (2016).
- [4] Bill Buxton. 2009. Mediaspace—meaningspace—meetingspace. In *Media space 20+ years of mediated life*. Springer, 217–231.
- [5] Cornelius Michael Deasy and Thomas E Lasswell. 1985. Designing places for people: A handbook on human behavior for architects, designers, and facility managers. (1985).
- [6] Maiken Hillerup Fogtmann, Peter Gall Krogh, and Thomas Markussen. 2011. Spatial sharing: Designing computational artifacts as architecture. In *Proceedings of the International Conference on Designing Pleasurable Products and Interfaces*. 1–9.
- [7] Jens Emil Grønbaek, Henrik Korsgaard, Marianne Graves Petersen, Morten Henriksen Birk, and Peter Gall Krogh. 2017. Proxemic transitions: designing shape-changing furniture for informal meetings. In *Proceedings of the 2017 CHI conference on human factors in computing systems*. 7029–7041.
- [8] Ge Guo, Gilly Leshed, and Keith Evan Green. 2023. "I normally wouldn't talk with strangers": Introducing a Socio-Spatial Interface for Fostering Togetherness Between Strangers. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (CHI '23). Association for Computing Machinery, New York, NY, USA, Article 272, 20 pages. doi:10.1145/3544548.3581325
- [9] Ge Guo, Gilly Leshed, Trevor Pinch, and Keith Evan Green. 2022. SocialStools: A Playful, Socio-Spatial Interface for Fostering Togetherness Across Strangers. In *Extended Abstracts of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA) (CHI EA '22). Association for Computing Machinery, New York, NY, USA, Article 173, 5 pages. doi:10.1145/3491101.3519877
- [10] Ge Guo, Qi Yang, Yunting Yan, Xiaoman Yang, Guy Hoffman, Gilly Leshed, and Keith Evan Green. 2025. MirrorBot: Exploring Socio-Spatial Interactions that Foster Serendipitous Human Connections Through Robotic Mirrors. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems* (CHI EA '24). ACM, Yokohama, Japan, 8 pages. doi:10.1145/3706599.3721176
- [11] Ge (Serena) Guo and Zhiwen Qiu. 2024. DentAR: Innovating Dental Visits with Sensory Experiences in AR for People with Autism Spectrum Disorder. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI EA '24). Association for Computing Machinery, New York, NY, USA, Article 99, 7 pages. doi:10.1145/3613905.3650926
- [12] Ge (Serena) Guo, Qi Yang, Rejoice Hu, and Guy Hoffman. 2025. Facilitating Synchronized Movement during Ice-Breaking Scenarios through a Real-World Reinforcement Learning Agent Using Non-Verbal Behaviors. In *Proceedings of the 2025 ACM/IEEE International Conference on Human-Robot Interaction* (Melbourne, Australia) (HRI '25). IEEE Press, 1324–1328. doi:10.5555/3721488.3721672
- [13] Edmund T Hall and Edward Twitchell Hall. 1966. *The Hidden Dimension*. Doubleday.
- [14] Keith N Hampton, Harrison Rainie, Weixu Lu, Maria Dwyer, Inyoung Shin, and Kristen Purcell. 2014. Social media and the 'spiral of silence'.
- [15] Luke Hespanhol, Martin Tomitsch, Oliver Bown, and Miriama Young. 2014. Using embodied audio-visual interaction to promote social encounters around large media façades. In *Proceedings of the 2014 conference on Designing interactive systems* (Vancouver, BC, Canada) (DIS '14). Association for Computing Machinery, New York, NY, USA, 945–954. doi:10.1145/2598510.2598568
- [16] Tim Ingold. 2000. Against Space: Place, Movement, Knowledge. In *The Perception of the Environment: Essays on Livelihood, Dwelling and Skill*, Tim Ingold (Ed.). Routledge, 219–242.
- [17] Viirj Kan, Katsuya Fujii, Judith Amores, Chang Long Zhu Jin, Pattie Maes, and Hiroshi Ishii. 2015. Social Textiles: Social Affordances and Icebreaking Interactions Through Wearable Social Messaging. In *Proceedings of the Ninth International Conference on Tangible, Embedded, and Embodied Interaction* (Stanford, California, USA) (TEI '15). Association for Computing Machinery, New York, NY, USA, 619–624. doi:10.1145/2677199.2688816
- [18] Adam Kendon. 1990. *Conducting interaction: Patterns of behavior in focused encounters*. Vol. 7. CUP Archive.
- [19] Seungki Kim, Jiwoo Hong, Jaeyeon Lee, Hyun-Sook Choi, Geehyuk Lee, and Woohun Lee. 2018. TouchBranch: Understanding Interpersonal Touches in Interactive Installation. In *Proceedings of the 2018 Designing Interactive Systems Conference* (Hong Kong, China) (DIS '18). Association for Computing Machinery, New York, NY, USA, 535–546. doi:10.1145/3196709.3196740
- [20] David Kirsh. 2019. Do Architects and Designers Think About Interactivity Differently? *ACM Transactions on Computer-Human Interaction (TOCHI)* 26, 1 (2019), Article 3. doi:10.1145/3301425
- [21] Ilpo Koskinen, John Zimmerman, Thomas Binder, Johan Redström, and Stephan Wensveen. 2011. *Design Research Through Practice: From the Lab, Field, and Showroom*. Elsevier.
- [22] Peter Gall Krogh, Marianne Graves Petersen, Kenton O'Hara, and Jens Emil Groenbaek. 2017. Sensitizing concepts for socio-spatial literacy in HCI. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*. 6449–6460.
- [23] Paul SN Lee, Louis Leung, Venhwei Lo, Chengyu Xiong, and Tingjun Wu. 2011. Internet communication versus face-to-face interaction in quality of life. *Social Indicators Research* 100 (2011), 375–389.
- [24] Henri Lefebvre. 1991. *The Production of Space*. Blackwell.
- [25] Nicolai Marquardt, Nathalie Henry Riche, Christian Holz, Hugo Romat, Michel Pahud, Frederik Brudy, David Ledo, Chunjong Park, Molly Jane Nicholas, Teddy Seyd, et al. 2021. AirConstellations: In-Air Device Formations for Cross-Device Interaction via Multiple Spatially-Aware Armatures. In *The 34th Annual ACM Symposium on User Interface Software and Technology*. 1252–1268.
- [26] Maurice Merleau-Ponty. 1962. *Phenomenology of Perception*. Routledge & Kegan Paul. Original work published 1945.
- [27] Tien T Nguyen, Duyen T Nguyen, Shamsi T Iqbal, and Eyal Ofek. 2015. The Known Stranger: Supporting Conversations between Strangers with Personalized Topic Suggestions. In *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems* (Seoul, Republic of Korea) (CHI '15). Association for Computing Machinery, New York, NY, USA, 555–564. doi:10.1145/2702123.2702411
- [28] Donald A. Norman. 1990. *The Design of Everyday Things*. MIT Press.

- [29] Antti Oulasvirta, Esko Kurvinen, and Tomi Kankainen. 2003. Understanding contexts by being there: case studies in bodystorming. *Personal and ubiquitous computing* 7 (2003), 125–134.
- [30] Peter Peltonen, Esko Kurvinen, Antti Salovaara, Giulio Jacucci, Tommi Ilmonen, John Evans, Antti Oulasvirta, and Petri Saarikko. 2008. It's Mine, Don't Touch! interactions at a large multi-touch display in a city centre. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Florence, Italy) (CHI '08). Association for Computing Machinery, New York, NY, USA, 1285–1294. doi:10.1145/1357054.1357255
- [31] Viva Sarah Press and Hadas Erel. 2023. Humorous Robotic Behavior as a New Approach to Mitigating Social Awkwardness. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (CHI '23, Article 427). Association for Computing Machinery, New York, NY, USA, 1–16. doi:10.1145/3544548.3580821
- [32] Ofir Sadka, Alon Jacobi, Andrey Grishko, Udi Lumnitz, Benny Megidish, and Hadas Erel. 2022. "By the Way, What's Your Name?": The Effect of Robotic Bar-Stools on Human-Human Opening-Encounters. In *Extended Abstracts of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA) (CHI EA '22). Association for Computing Machinery, New York, NY, USA, Article 331, 6 pages. doi:10.1145/3491101.3519726
- [33] Elizabeth B-N Sanders and Pieter Jan Stappers. 2008. Co-creation and the new landscapes of design. *Co-design* 4, 1 (2008), 5–18.
- [34] Edward W. Soja. 1996. *Thirdspace: Journeys to Los Angeles and Other Real-and-Imagined Places*. Blackwell.
- [35] Maurício Sousa, Daniel Mendes, Alfredo Ferreira, João Madeiras Pereira, and Joaquim Jorge. 2015. Eery space: facilitating virtual meetings through remote proxemics. In *Human-Computer Interaction-INTERACT 2015: 15th IFIP TC 13 International Conference, Bamberg, Germany, September 14-18, 2015, Proceedings, Part III* 15. Springer, 622–629.
- [36] Maurício Sousa, Daniel Mendes, Daniel Medeiros, Alfredo Ferreira, João Madeiras Pereira, and Joaquim Jorge. 2016. Remote proxemics. *Collaboration meets interactive spaces* (2016), 47–73.
- [37] Moonyoung Tae and Joonhwan Lee. 2020. The Effect of Robot's Ice-breaking Humor on Likeability and Future Contact Intentions. In *Companion of the 2020 ACM/IEEE International Conference on Human-Robot Interaction* (Cambridge, United Kingdom) (HRI '20). Association for Computing Machinery, New York, NY, USA, 462–464. doi:10.1145/3371382.3378267
- [38] Yi-Fu Tuan. 1977. *Space and Place: The Perspective of Experience*. University of Minnesota Press.
- [39] Sherry Turkle. 2017. *Alone together: Why we expect more from technology and less from each other*. Hachette UK.
- [40] Brygg Ullmer and Hiroshi Ishii. 2000. Emerging Frameworks for Tangible User Interfaces. *IBM Systems Journal* 39, 3.4 (2000), 915–931.
- [41] Alex Wuqi Zhang, Ting-Han Lin, Xuan Zhao, and Sarah Sebo. 2023. Ice-Breaking Technology: Robots and Computers Can Foster Meaningful Connections between Strangers through In-Person Conversations. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (CHI '23, Article 273). Association for Computing Machinery, New York, NY, USA, 1–14. doi:10.1145/3544548.3581135
- [42] John Zimmerman, Jodi Forlizzi, and Shelley Evenson. 2007. Research Through Design as a Method for Interaction Design Research in HCI. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. 493–502. doi:10.1145/1240624.1240704